



# Professional audio and voice solutions

**Eduardo Gonzalez Reyes**

November 2025

# Differentiated Voice solutions from NXP

1

Voice for MCU – Wearable  
i.MX RT 3 digits

## Differentiation

- Low Power MCU
- Voice & Audio AI Software solution for wearables

2

Voice standard enablement  
for i.MX Voice Plugin

## Differentiation

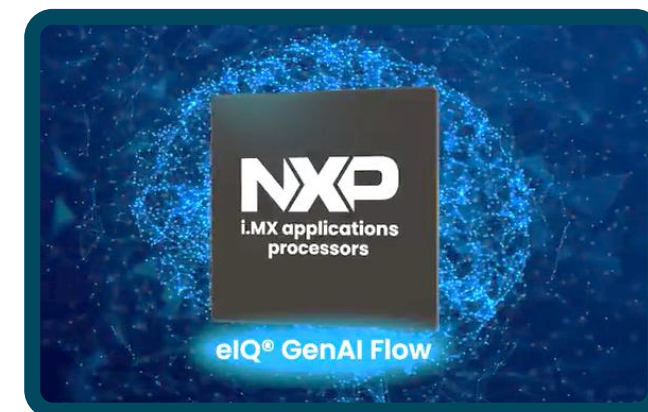
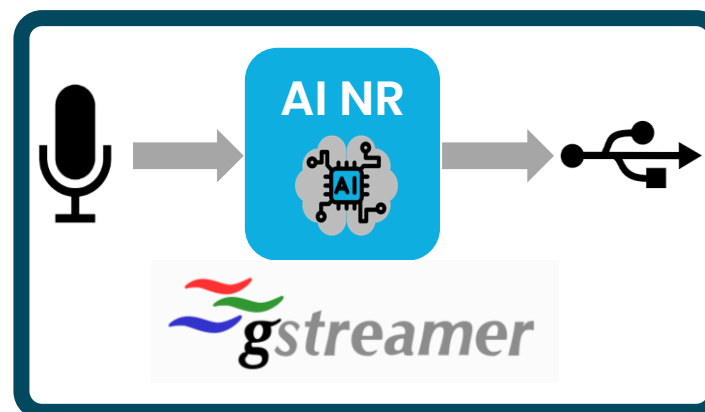
- Ease of integration
- Audio AI brings good performance and 0 tuning

3

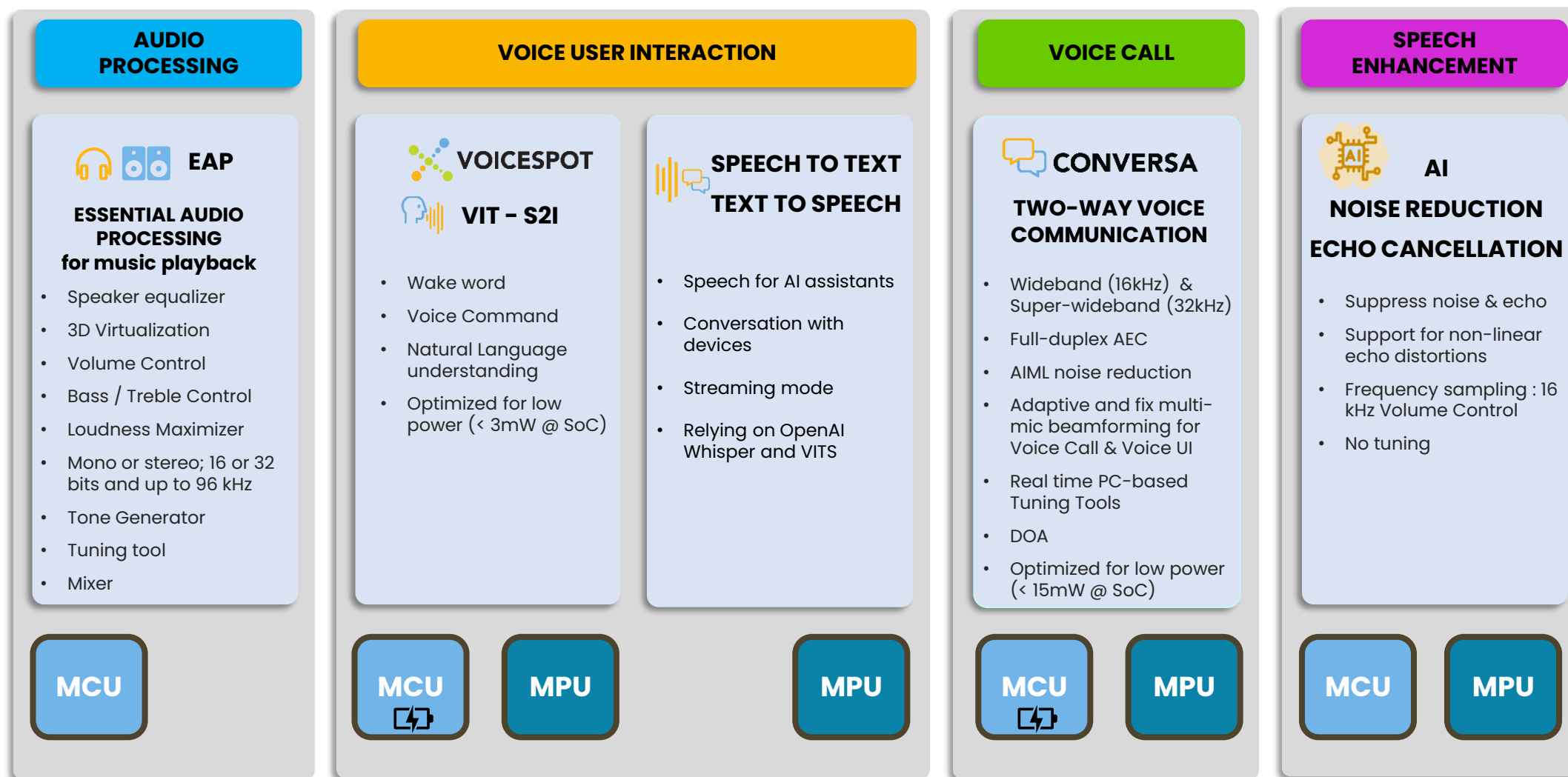
Generative AI for i.MX

## Differentiation

- Generative AI on Edge
- Enable ASR/LLM/TTS with i.MX95 Neutron
- Enable ASR/VLM/TTS on Edge with Kinara



# Voice & Audio Portfolio



# Voice & Audio software products



# Conversa – full duplex voice communication software

## APPLICATIONS

Wearables, Smart Watches, Security Cameras, Doorbells, Intercoms, Gaming Headsets

## PLATFORMS

i.MX RT500 / RT600 / RT700  
i.MX8, i.MX9

## Conversa features

### Conversa Rx Tx features

#### Path Rx:

- 1 input / output channel
- High pass filter (HPF)
- Noise reduction (NR)
- Equalizer (4 curves, can support more)
- Dynamic compressor (2 sub bands + 1 full band + 1 limiter)
- Gain (In/Out)

#### Path Tx:

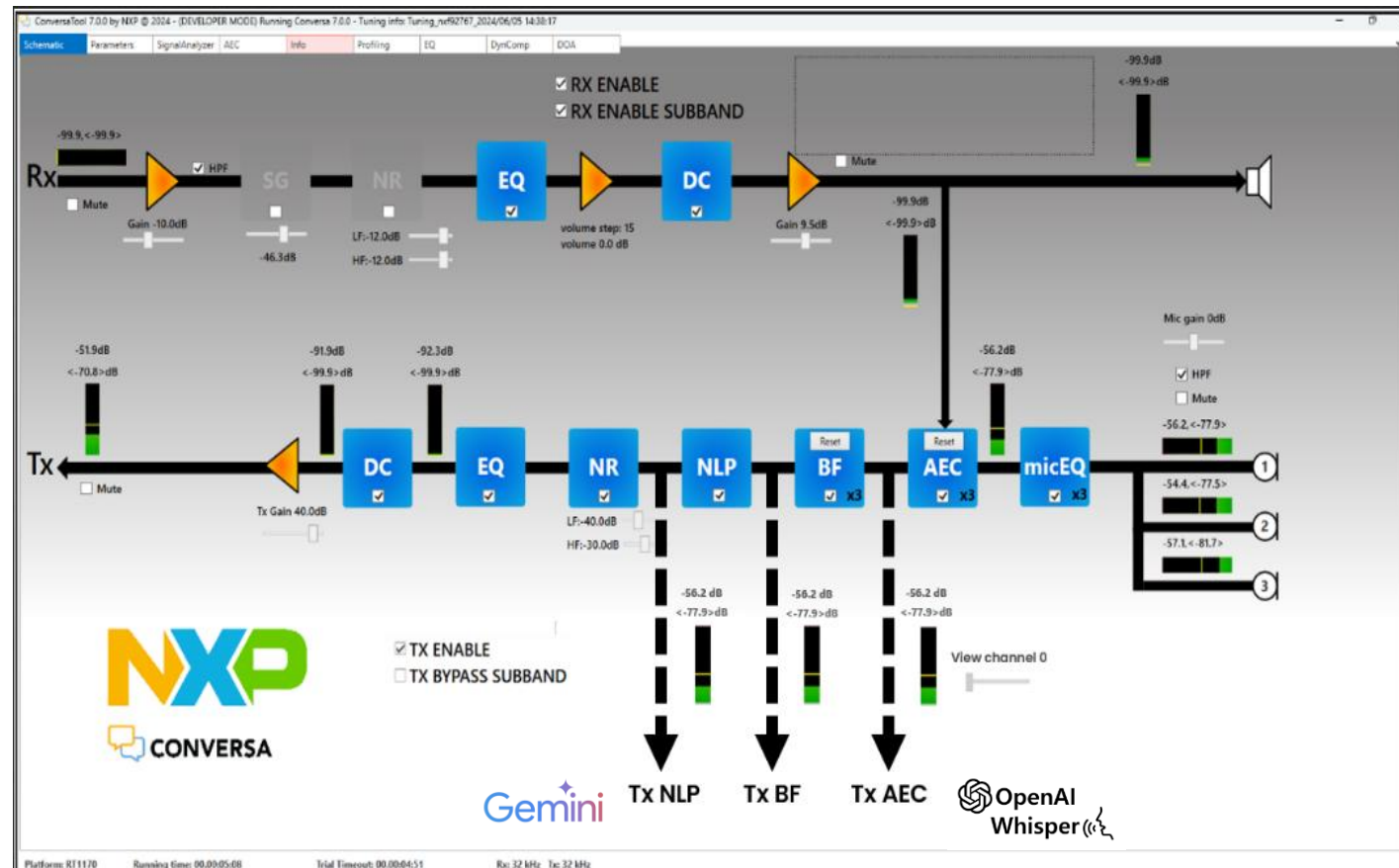
- 1 to 4 microphones input channel
- 1 output channel
- High pass filter (HPF)
- Acoustic echo canceller (AEC - 17ms)
- Adaptive and Fix Beam former
- Noise suppressor (AI)
- Equalizer (4 curves, can support more)
- Dynamic compressor (2 sub bands + 1 full band + 1 limiter)
- Gain (In/Out)

#### Audio stream :

- 16kHz => frame size = 8ms / 128 samples
- 32kHz => frame size = 4ms / 128 samples

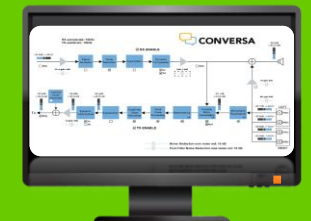
#### Low Power :

- Can be optimized for Low Power on i.MXRT500/600/700



## Engineering services

- Audio / Acoustic Design
- Testing, Tuning & Debug
- System Engineering & Integration
- Microsoft Teams Precertification



# AI Acoustic Echo Canceller & Noise Reduction

## APPLICATIONS

Speech denoising with optional echo removal for full duplex

- End-to-End deep learning system
- No tuning required



### AI-NR : Noise Reduction [demo](#)

- Frequency sampling : 16 kHz & 32 kHz
- Suppress noise in the microphone signal
- De-reverberation

### AI-AECNR Acoustic Echo Canceller & Noise Reduction : [demo](#)

- Frequency sampling : 16 kHz
- Suppress echo and noise in the microphone signal
- Support for non-linear distortions
- Support variable echo path delay

## PLATFORMS

i.MX RT500 / RT600 / RT700 (AI-NR only)  
i.MX8, i.MX9 with i.MX Voice plugin through GStreamer

### AI-NR : Noise Reduction

| Model      | Runtime memory [kB] | Sample rate [kHz] | Core              | CPU load [%] |
|------------|---------------------|-------------------|-------------------|--------------|
| Small 50k  | 4.6 kB              | 16                | FusionF1 @ 240MHz | 2.8%         |
|            |                     |                   | HiFi4 @ 600 MHz   | 0.8%         |
|            |                     | 32                | FusionF1 @ 240MHz | 5.6%         |
|            |                     |                   | HiFi4 @ 600 MHz   | 1.6%         |
| Medium 72k | 38 kB               | 16                | FusionF1 @ 240MHz | 11.4%        |
|            |                     |                   | HiFi4 @ 600 MHz   | 3.5%         |
|            |                     | 32                | FusionF1 @ 240MHz | 22.8%        |
|            |                     |                   | HiFi4 @ 600 MHz   | 7%           |

### AI-AECNR : Acoustic Echo Canceller & Noise Reduction

| Model      | Sample rate [kHz] | Core                   | CPU load [%] | Weight format |
|------------|-------------------|------------------------|--------------|---------------|
| Small 200k | 16 kHz            | 1x Cortex-A55 @ 1.7GHz | 23           | FP32          |
| Large 600k | 16 kHz            | 1x Cortex-A55 @ 1.7GHz | 43           | FP32          |

# Voice User Interface enablement

|                                  | Wake word                        |                                 | Voice Command                            | Natural Language Understanding                                                     |
|----------------------------------|----------------------------------|---------------------------------|------------------------------------------|------------------------------------------------------------------------------------|
|                                  | VoiceSpot                        | VIT                             | VIT Voice Command                        | VIT Speech to Intent                                                               |
| <b>Function</b>                  | Wake Word Engine                 |                                 | Up to 30 voice commands                  | NLU for controlling a device                                                       |
| <b>Best For</b>                  | Highest Performance<br>Low-Power | Easily customized<br>wake-words | Basic voice control with simple commands | Natural Language Voice UI, supports many<br>expressions or ways of saying commands |
| <b>Online Tools</b>              | No                               | Yes                             | Yes                                      | Yes                                                                                |
| <b>Training Corpus Required?</b> | Yes                              | No                              | No                                       | No                                                                                 |
| <b>Other Considerations</b>      | Training corpus required         | Customer self-service<br>models | Customer self-service models             | NXP creates models for customer                                                    |
| <b>Availability</b>              | Controlled Release               | SDK, BSP                        | SDK, BSP                                 | Controlled Release                                                                 |
| <b>How to Get Started</b>        | voice@nxp.com                    | www.nxp.com/vit                 | www.nxp.com/vit                          | voice@nxp.com                                                                      |
| <b>Cost</b>                      | Premium                          | Standard                        | Standard                                 | Premium                                                                            |

# VoiceSpot : Local wake word

## APPLICATIONS

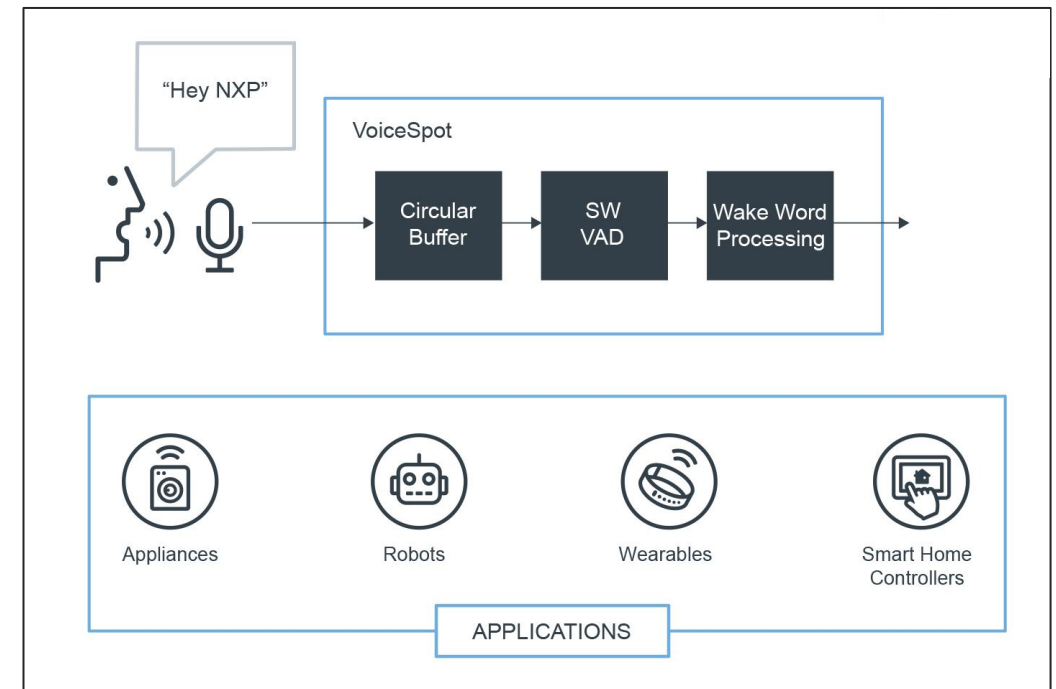
Ultra Low Power, Voice UI enablement

- Small memory footprint, <100kB including model, heap & program memory
- Low computational complexity allows always on functionality
- Average Power on i.MX RT700 / HiFi1 : 1.2 mW
- Requires a training corpus training database (~ \$30K / wake word)

| Platform                    | Full-Speed Wake Word Processing |
|-----------------------------|---------------------------------|
| i.MX RT500 / Cadence Fusion | 8 MHz                           |
| i.MX RT600 / Cadence Hifi4  | 6.5 MHz                         |
| i.MX RT700 / Cadence Hifi1  | 13 MHz                          |

## PLATFORMS

i.MX RT500, i.MX RT600, i.MX RT700





# VIT : Local voice wake word voice commands and intent detection

## APPLICATIONS

Voice UI enablement with audio front-end, wake word detection and voice command or intent detection

## PLATFORMS

LPC55S69, MCXN94, i.MXRT500/600/700, i.MXRT1040 / 1050 / 1060 / 1160 / 1170, RW610, i.MX8M Mini, i.MX8M Nano, i.MX8M Plus, i.MX93, i.MX95.

[VIT Cloud Model generation tool](#): provide the ability to customers to build their own model without training database

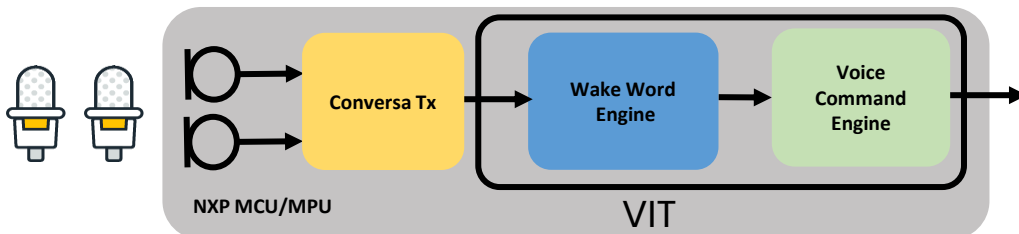
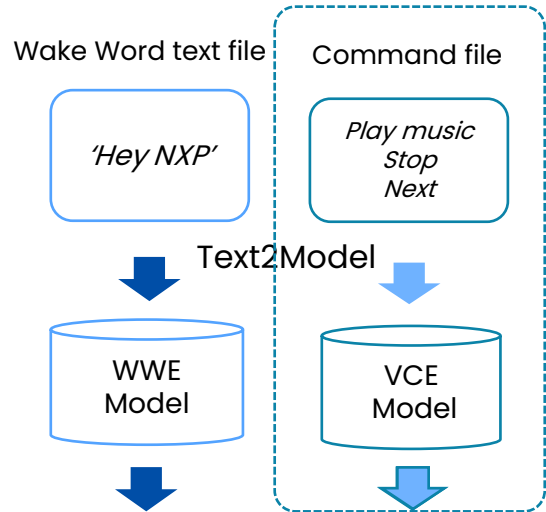
### Command :

Play Music  
Stop

[Smart Watch i.MX RT500](#)

[Smart Aircon RW612](#)

[Smart Oven i.MX RT1170](#)

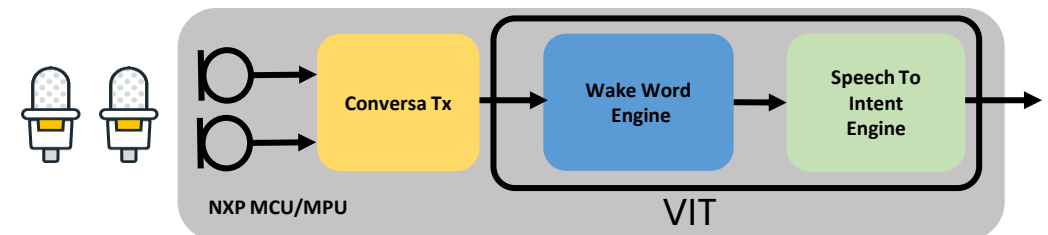
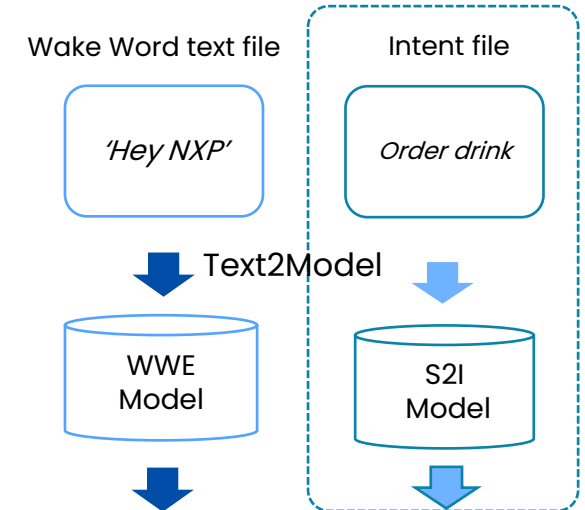


Option 1 : VIT Wake word + **Voice command**

**Intent:** "Can I have a single espresso with some sugar?"

Intent : **Order Drink**

| Slot Tag | Slot value      |
|----------|-----------------|
| Drink    | <b>espresso</b> |
| size     | <b>single</b>   |
| sugar    | <b>some</b>     |



Option 2 : VIT Wake word + **Speech To Intent**

# Speech to Text & Text To Speech

## APPLICATIONS

Powering speech for AI assistants  
Enabling conversation with devices

## PLATFORMS

Speech To Text : i.MX8MP, i.MX93, i.MX95  
Text To Speech : i.MXRT700, i.MX8MP, i.MX93, i.MX95

### Speech To Text : [demo](#)

- Streaming mode
- Faster than any open-source package
- Leveraging Whisper Encoder 8bits on Neutron NPU
- Q8 Weights quantization
- ONNX RT inference engine



| Model             | Model size   | WER                           | Avg (Min/Max) latency time every 3 sec of a 30 sec speech streaming on 6x Cortex-A55 @ 2.0GHz |
|-------------------|--------------|-------------------------------|-----------------------------------------------------------------------------------------------|
|                   | [parameters] | [%] on LibriSpeech test-clean | [second]                                                                                      |
| Whisper-tiny.en   | 39M          | 12.2                          | 0.78 (0.31/1.37)                                                                              |
| Whisper-base.en   | 74M          | 6.96                          | 1.26 (0.49/2.16)                                                                              |
| Whisper-small.en  | 244M         | 3.98                          | 3.02 (1.22/4.45)                                                                              |
| Whisper-medium.en | 769M         | 6.55                          | 10.82 (4.24/14.42)                                                                            |

### Text To Speech : [demo](#)



Small TTS for MCU i.MXRT700 NPU



Large TTS streaming for MPU

| Model                                       | DNS-MOS    | model size |
|---------------------------------------------|------------|------------|
| Real speech (reference)                     | 4.39       | -          |
| Small : MCU LightSpeech + HiFiGAN model     | 4.17 ±0.22 | 2.2M       |
| Large : MPU Vits model female/ male speaker | 4.40 ±0.15 | 20M        |

# EAP : Essential Audio Processing for stereo audio playback

## APPLICATIONS

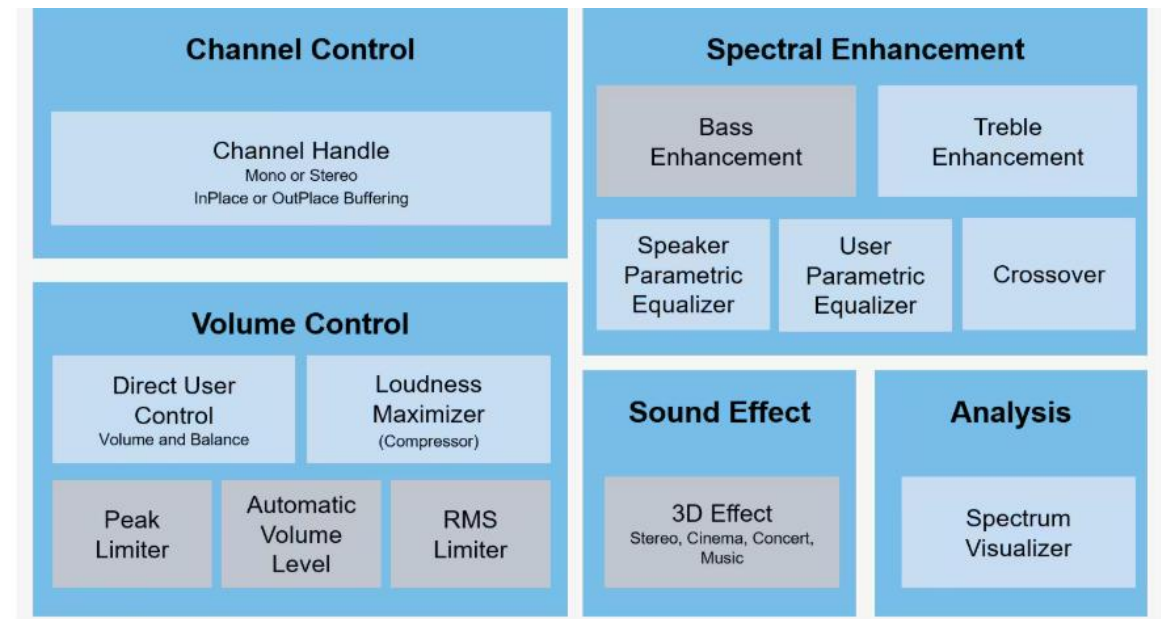
Music player, stereo audio processing for playback

Free, ready-to-use library that provides a complete set of mono and stereo audio playback control features.

- Tone Generator
- Speaker equalizer
- 3D Virtualization: Concert Sound
- Volume Control
- User Equalizer
- Bass Enhancement
- Auto Volume Leveler
- Loudness Maximizer
- Treble Enhancement
- Peak Limiter
- RMS Limiter
- Parametric Spectrum Analyzer
- Crossover 2-Bands
- Supports mono or stereo raw PCM
- 16 or 32 bits and up to 96 kHz
- Tuning tool provided for EAP tuning
- Audio Mixer

## PLATFORMS

i.MX RT500 / RT600 / RT700

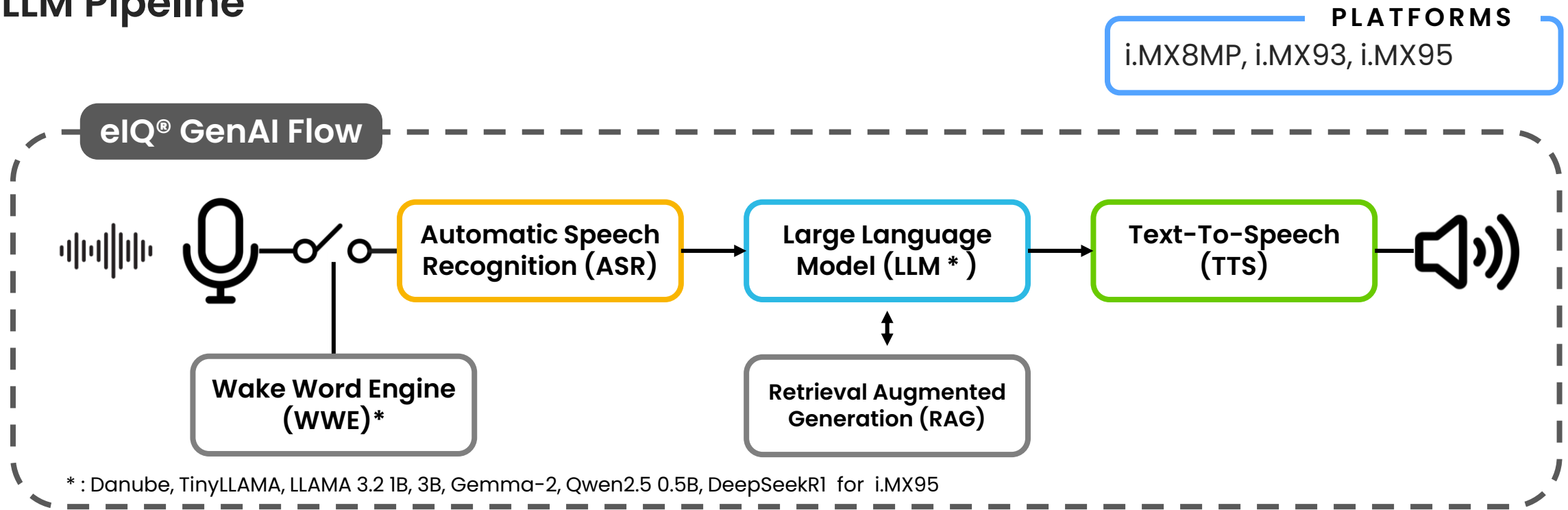


**Lib EAP16** : Mono & Stereo / 16-bits/ up to 48 kHz

**Lib EAP32** : Mono & Stereo /24 &32 bits/ up to 96kHz

<https://nxp.com/eap>

# LLM Pipeline



## NXP differentiation

- **Faster ASR** than any open-source alternative
- **LLM offloaded on i.MX95 Neutron NPU**
- **Human-inspired RAG** for better LLM fine-tuning for customer
- **Customize Gen AI flow** : New languages, Audio Front End, DOA

Available on [Github](#)



eIQ Gen AI Video demo [here](#)

# Immersiv3D

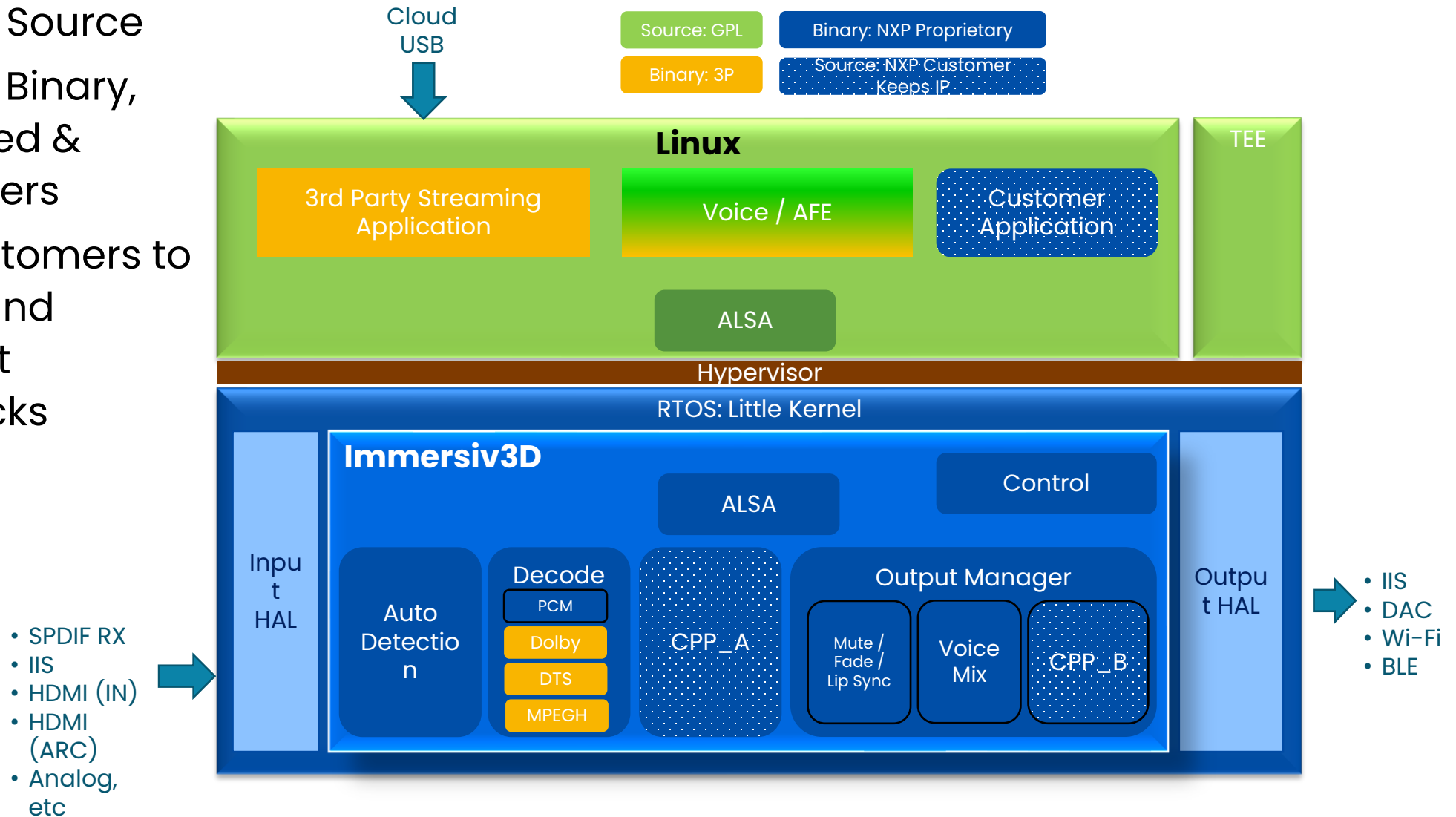


# Audio Product Target Markets



# Immersiv3D Architecture

- BSP provided in Source
- I3D Provided as Binary, includes Licensed & Certified Decoders
- API enables customers to add standard and proprietary post processing blocks





# Immersiv3D Release Contents

1. **imx8mm\_I3D\_binaries.tar.gz**: Immersiv3D pre-built binary (lk.bin), jailhouse modified binary and configuration files, and Linux Kernel modules.
2. **imx8mm\_I3D\_metalayer.tar.gz**: Customer metalayers with includes patches, recipes and Immersiv3D SDK (meta-audio-framework).
3. **image-imx8mmab2af.sdcard.bz2**: Prebuilt sdcard image that contains all the platform installed and sample Post Processing Application provided for demonstration.
4. **imx-boot-imx8mmab2af-sd.bin-flash\_evk.bz2**: Prebuilt boot image containing bootloader binary
5. **LICENSE.txt**: LA\_OPT\_NXP\_Software\_License

## Software Files

## Documentation

1. **Immersiv3D\_AppNote\_V3\_EAR1.0.0.pdf**: App Note to help with development and porting of audio post processing algorithms to the i.MX 8M Mini Immersiv3D audio framework.
2. **Immersiv3D\_UserGuide\_V3\_EAR1.0.0.pdf**: Provide details on setup and usage of the release with the i.MX 8M mini EVK.
3. **Release\_Notes\_imx8m\_Immersiv3D\_Version3\_EAR1.0.0.pdf**: This document describing the release.
4. **EULA.txt**: NXP LA\_OPT\_NXP\_Software\_License Software License Agreement
5. **SCR-5.15.5-1.0.0.txt**: NXP Software Content Register for Linux 5.15.5-1.0.0 version.
6. **SW-Content-Register-imx8mm\_I3D\_SDK\_V3EAR1.0.0.txt**: NXP Software Content Register for i.MX8M Mini package.

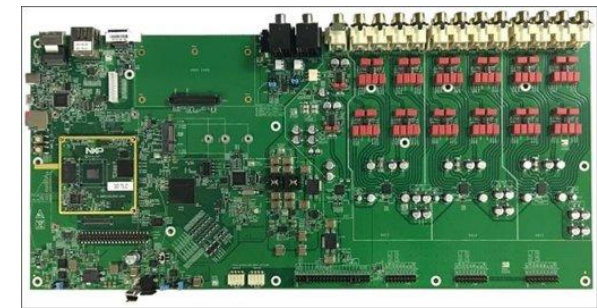
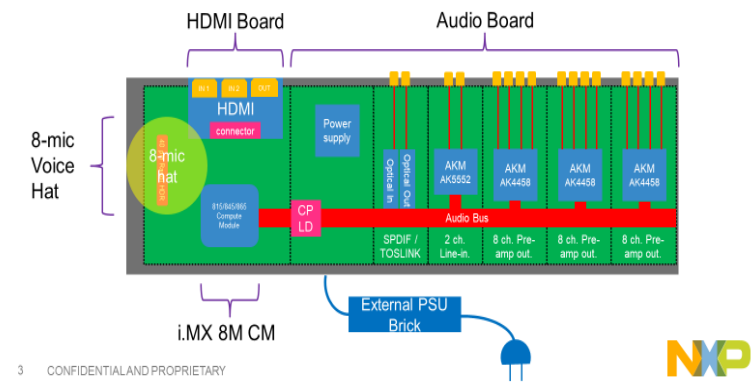


# Audio HW Enablement and Orderable Part Numbers | i.MX Audio Board

| Part Number | Features                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | S/R Price (USD) |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| MCIMX8M-AUD | <ul style="list-style-type: none"><li>i.MX Audio Board supports i.MX 8M Plus, i.MX 8M Mini &amp; i.MX 8M Nano compute modules/SOMs (Mini and Nano require interposer)</li><li>Includes :<ul style="list-style-type: none"><li>24 Output Channels @ 192KHz</li><li>2 Input Channels @ 192KHz</li><li>Ethernet</li><li>PCIe</li><li>USB</li><li>Raspberry Pi Header</li><li>Encloser with Buttons</li><li>i.MX 8M Mini &amp; i.MX 8M Nano interposer for Audio Board 2.0</li><li>HDMI Card</li></ul></li></ul> | \$1,840         |

Product page with “Buy Direct” link:  
<https://www.nxp.com/part/MCIMX8M-AUD - />

Complete Four Board System in Chassis Block Diagram



# Audio HW Enablement and Orderable Part Numbers | i.MX 8M Mini

|                               | Features                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | S/R Price (USD) |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <a href="#">M845I3DCA-CPU</a> | <ul style="list-style-type: none"><li>• Compute module part number that includes “MIMX8MM5DVT LZCA” processor. <b>This compute module replaces the standard compute module to support Dolby Atmos.</b></li><li>• Includes the same configuration as the “8MMINILPD4-EVK” compute module:<ul style="list-style-type: none"><li>– LPDDR4 memory</li><li>– Murata Wi-Fi &amp; BT module (Qualcomm QCA9377 chipset)</li><li>– eMMC</li></ul><a href="https://www.nxp.com/part/M845I3DCA-CPU - /">https://www.nxp.com/part/M845I3DCA-CPU - /</a></li></ul>       | \$208.95        |
| <a href="#">M845I3DDA-CPU</a> | <ul style="list-style-type: none"><li>• Compute module part number that includes “MIMX8MM5DVT LZDA” processor. <b>This compute module replaces the standard compute module to support Dolby Atmos+DTS:X.</b></li><li>• Includes the same configuration as the “8MMINILPD4-EVK” compute module:<ul style="list-style-type: none"><li>– LPDDR4 memory</li><li>– Murata Wi-Fi &amp; BT module (Qualcomm QCA9377 chipset)</li><li>– eMMC</li></ul><a href="https://www.nxp.com/part/M845I3DDA-CPU - /">https://www.nxp.com/part/M845I3DDA-CPU - /</a></li></ul> | \$208.95        |

Compute Module



# Your feedback is important to us!

Please fill out the session survey for this class using the QR code or by clicking into the class within the app.





# Get in touch

**Eduardo Gonzales Reyes**

eduardo.gonzalez@nxp.com

[nxp.com](https://www.nxp.com)



[nxp.com](https://nxp.com)

**| Public |** NXP and the NXP logo are trademarks of NXP B.V. All other product or service names are the property of their respective owners. © 2025 NXP B.V. Version 2.9.